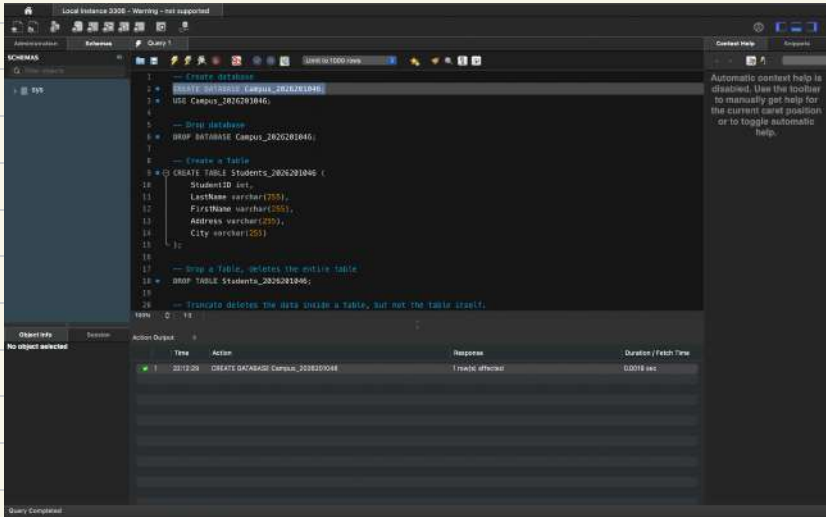
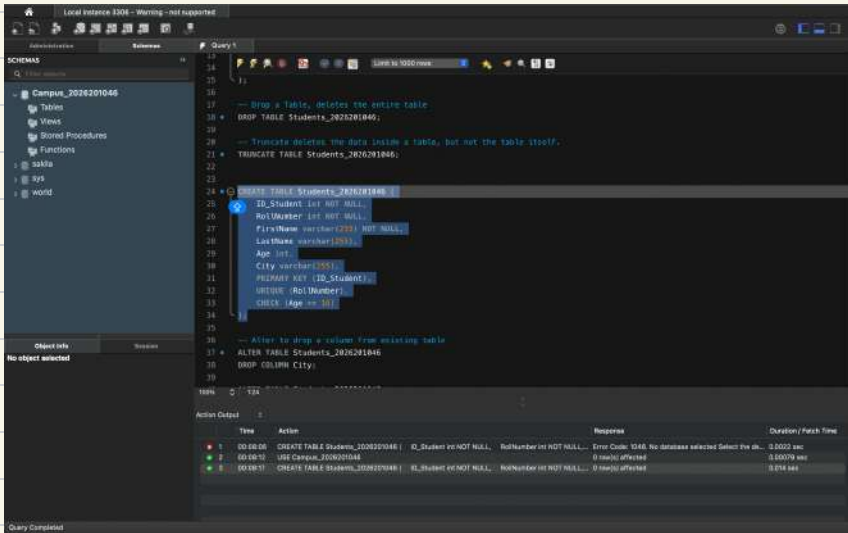


1. Create Database



2. Create a table



3. Drop Database

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1 -- Create database
2 * CREATE DATABASE Campus_2026201046;
3 * USE Campus_2026201046;
4
5 -- Drop database
6 * DROP DATABASE Campus_2026201046;
7
8
9 * -- Create a Table
10 CREATE TABLE Students_2026201046 (
11     StudentID int,
12     LastName varchar(255),
13     FirstName varchar(255),
14     Address varchar(255),
15     City varchar(255)
16 );
17
18 -- Drop a Table, deletes the entire table
19 * DROP TABLE Students_2026201046;
20
21 -- Truncate deletes the data inside a table, but not the table itself.
22 * TRUNCATE TABLE Students_2026201046;
```

The 'Action Output' pane shows the execution results:

Time	Action	Response	Duration / Fetch Time
22:12:29	CREATE DATABASE Campus_2026201046	1 row(s) affected	0.0219 sec
22:12:47	USE Campus_2026201046	0 row(s) affected	0.00096 sec
22:12:51	CREATE TABLE Students_2026201046 (StudentID int, LastName varchar(255),	0 row(s) affected	0.070 sec
22:13:08	DROP DATABASE Campus_2026201046	1 row(s) affected	0.016 sec

4. Drop a table

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window contains the following SQL script:

```
1
2
3
4
5
6
7
8
9 * -- Create TABLE Students_2026201046 (
10 StudentID int,
11 LastName varchar(255),
12 FirstName varchar(255),
13 Address varchar(255),
14 City varchar(255)
15 );
16
17 -- Drop a Table, deletes the entire table
18 * DROP TABLE Students_2026201046;
19
20 -- Truncate deletes the data inside a table, but not the table itself.
21 * TRUNCATE TABLE Students_2026201046;
22
23 -- Final working definition of Students_2026201046, built up with constraints below.
24 -- (Each CREATE TABLE below was previously named "Students"/"Person" incrementally,
25 -- and even referenced a non-existent "Person" table. Consolidated & fixed here.)
26
27 * -- Create TABLE Students_2026201046 (
28     ID_Student int NOT NULL,
29     ...
30 );
```

The 'Action Output' pane shows the execution results:

Time	Action	Response	Duration / Fetch Time
22:12:29	CREATE DATABASE Campus_2026201046	1 row(s) affected	0.0219 sec
22:12:47	USE Campus_2026201046	0 row(s) affected	0.00096 sec
22:12:51	CREATE TABLE Students_2026201046 (StudentID int, LastName varchar(255),	0 row(s) affected	0.070 sec
22:12:58	DROP DATABASE Campus_2026201046	1 row(s) affected	0.016 sec
22:13:08	CREATE DATABASE Campus_2026201046	1 row(s) affected	0.0029 sec
22:14:07	USE Campus_2026201046	0 row(s) affected	0.00049 sec
22:14:14	CREATE TABLE Students_2026201046 (StudentID int, LastName varchar(255),	0 row(s) affected	0.0061 sec
22:14:21	DROP TABLE Students_2026201046	0 row(s) affected	0.0042 sec

5. Alter to drop a column from existing table

The screenshot shows a SQL query in a text editor. The query is as follows:

```
1 Firstname varchar(255) NOT NULL,
2 LastName varchar(255),
3 Age int,
4 City varchar(255),
5 PRIMARY KEY (ID_Student),
6 UNIQUE (RollNumber),
7 CHECK (Age >= 18)
8 );
9
10 -- Alter to drop a column from existing table
11 ALTER TABLE Students_2026201046
12 DROP COLUMN City;
13
14 ALTER TABLE Students_2026201046
15 ADD COLUMN City varchar(255);
16
17 -- Alter the column definition of a column from existing table
18 ALTER TABLE Students_2026201046
19 MODIFY COLUMN LastName varchar(300);
20
```

The query is executed, and the results are shown in the bottom panel. The results table has columns: Time, Action, Response, and Duration / Patch Time. The actions performed are:

- TRUNCATE TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- TRUNCATE TABLE Students_2026201046
- DROP TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- ALTER TABLE Students_2026201046 DROP CO...
- ALTER TABLE Students_2026201046 ADD COL...
- ALTER TABLE Students_2026201046 MODIFY...
- DROP TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- ALTER TABLE Students_2026201046 DROP CO...

The screenshot shows a SQL query in a text editor. The query is as follows:

```
1 Firstname varchar(255) NOT NULL,
2 LastName varchar(255),
3 Age int,
4 City varchar(255),
5 PRIMARY KEY (ID_Student),
6 UNIQUE (RollNumber),
7 CHECK (Age >= 18)
8 );
9
10 -- Alter to drop a column from existing table
11 ALTER TABLE Students_2026201046
12 DROP COLUMN City;
13
14 ALTER TABLE Students_2026201046
15 ADD COLUMN City varchar(255);
16
17 -- Alter the column definition of a column from existing table
18 ALTER TABLE Students_2026201046
19 MODIFY COLUMN LastName varchar(300);
20
```

The query is executed, and the results are shown in the bottom panel. The results table has columns: Time, Action, Response, and Duration / Patch Time. The actions performed are:

- TRUNCATE TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- TRUNCATE TABLE Students_2026201046
- DROP TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- ALTER TABLE Students_2026201046 DROP CO...
- ALTER TABLE Students_2026201046 ADD COL...
- ALTER TABLE Students_2026201046 MODIFY...
- DROP TABLE Students_2026201046
- CREATE TABLE Students_2026201046 (ID_...)
- ALTER TABLE Students_2026201046 DROP CO...

7. Alter the column definition of a column from existing table

```

30  CREATE TABLE Students_2020201046 (
31  FirstName varchar(255) NOT NULL,
32  LastName varchar(255),
33  Age int,
34  City varchar(255),
35  PRIMARY KEY (ID_Student),
36  UNIQUE (RollNumber),
37  CHECK (Age >= 16)
38  );
39
40  -- Alter to drop a column from existing table
41  ALTER TABLE Students_2020201046
42  DROP COLUMN City;
43
44  -- Alter to add a column from existing table
45  ALTER TABLE Students_2020201046
46  ADD COLUMN City varchar(255);
47
48  -- Alter the column definition of a column from existing table
49  ALTER TABLE Students_2020201046
50  MODIFY COLUMN LastName varchar(500);
51
52

```

Time	Action	Response	Duration / Fetch Time
10:22:14.89	CREATE TABLE Students_2020201046 (ID_Student int NOT NULL, LastName varchar(255) NOT NULL, Age int, City varchar(255), PRIMARY KEY (ID_Student), UNIQUE (RollNumber), CHECK (Age >= 16));	0 rows(s) affected	0.013 sec
10:22:15.14	TRUNCATE TABLE Students_2020201046	0 rows(s) affected	0.012 sec
10:22:15.28	DROP TABLE Students_2020201046	0 rows(s) affected	0.0038 sec
10:22:15.34	CREATE TABLE Students_2020201046 (ID_Student int NOT NULL, LastName varchar(255) NOT NULL, Age int, City varchar(255), PRIMARY KEY (ID_Student), UNIQUE (RollNumber), CHECK (Age >= 16));	0 rows(s) affected	0.006 sec
10:22:15.57	ALTER TABLE Students_2020201046 DROP COLUMN City;	0 rows(s) affected	0.002 sec
10:22:15.60	ALTER TABLE Students_2020201046 ADD COLUMN City varchar(255);	0 rows(s) affected	0.001 sec
10:22:15.72	ALTER TABLE Students_2020201046 MODIFY COLUMN LastName varchar(500);	0 rows(s) affected	0.0002 sec
10:22:15.79	ALTER TABLE Students_2020201046 DROP COLUMN City;	0 rows(s) affected	0.002 sec
10:22:15.88	DROP TABLE Students_2020201046	0 rows(s) affected	0.001 sec
10:22:15.94	CREATE TABLE Students_2020201046 (ID_Student int NOT NULL, LastName varchar(500) NOT NULL, Age int, City varchar(255), PRIMARY KEY (ID_Student), UNIQUE (RollNumber), CHECK (Age >= 16));	0 rows(s) affected	0.0071 sec
10:22:15.99	ALTER TABLE Students_2020201046 DROP COLUMN City;	0 rows(s) affected	0.002 sec
10:22:16.04	ALTER TABLE Students_2020201046 ADD COLUMN City varchar(255);	0 rows(s) affected	0.001 sec
10:22:16.11	ALTER TABLE Students_2020201046 MODIFY COLUMN LastName varchar(500);	0 rows(s) affected	0.0002 sec

8. Constraints by altering the table with NOT NULL

```

30  CREATE TABLE Students_2020201046 (
31  PRIMARY KEY (ID_Student),
32  UNIQUE (RollNumber),
33  CHECK (Age >= 16)
34  );
35
36  -- Alter to drop a column from existing table
37  ALTER TABLE Students_2020201046
38  DROP COLUMN City;
39
40  -- Alter to add a column from existing table
41  ALTER TABLE Students_2020201046
42  ADD COLUMN City varchar(255);
43
44  -- Alter the column definition of a column from existing table
45  ALTER TABLE Students_2020201046
46  MODIFY COLUMN LastName varchar(500);
47
48  -- Constraints by altering the table with NOT NULL
49  ALTER TABLE Students_2020201046
50  MODIFY COLUMN Age int NOT NULL;
51
52  -- Foreign Key Constraint
53  CREATE TABLE Subject_2020201046 (
54  SubjectID int NOT NULL,
55  PK_ID_Student int NOT NULL,
56  SubjectName varchar(255) NOT NULL,
57  FOREIGN KEY (PK_ID_Student) REFERENCES Students_2020201046 (ID_Student)
58  );
59

```

Time	Action	Response	Duration / Fetch Time
10:26:08	CREATE TABLE Students_2020201046 (ID_Student int NOT NULL, LastName varchar(255) NOT NULL, Age int, City varchar(255), PRIMARY KEY (ID_Student), UNIQUE (RollNumber), CHECK (Age >= 16));	Error Code: 1046, No database selected Select the db...	0.0022 sec
10:26:10	USE Campus_2020201046	0 rows(s) affected	0.00019 sec
10:26:11	CREATE TABLE Students_2020201046 (ID_Student int NOT NULL, LastName varchar(255) NOT NULL, Age int, City varchar(255), PRIMARY KEY (ID_Student), UNIQUE (RollNumber), CHECK (Age >= 16));	0 rows(s) affected	0.016 sec
10:26:12	ALTER TABLE Students_2020201046 DROP COLUMN City;	0 rows(s) affected	0.0002 sec

9. Foreign key constraint

The screenshot shows the SQL Studio interface with a query window containing the following SQL code:

```
ALTER TABLE Students_2020201040
MODIFY COLUMN LastName varchar(500);

-- Constraints by altering the table with NOT NULL
ALTER TABLE Students_2020201040
MODIFY COLUMN Age int NOT NULL;

-- Foreign Key Constraint
CREATE TABLE Subject_2020201040 (
  SubjectID int NOT NULL,
  PK_ID_Student int NOT NULL,
  SubjectName varchar(255) NOT NULL,
  No/Number int,
  Marks int,
  Semester int,
  PRIMARY KEY (SubjectID),
  CONSTRAINT PK_ID_Student FOREIGN KEY (PK_ID_Student)
  REFERENCES Students_2020201040 (ID_Student);
```

The Action Output pane at the bottom shows the execution results of the queries:

Time	Action	Response	Duration / Fetch Time
00:00:00	CREATE TABLE Students_2020201040 (ID_Student int NOT NULL, No/Number int NOT NULL, ...)	Error Code: 1046, No database selected Select the da...	0.0032 sec
00:00:02	USE Campus_2020201040	0 row(s) affected	0.00079 sec
00:00:02	CREATE TABLE Subject_2020201040 (ID_Student int NOT NULL, No/Number int NOT NULL, ...)	0 row(s) affected	0.014 sec
00:00:02	ALTER TABLE Students_2020201040 MODIFY COLUMN Age int NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.003 sec
00:00:07	CREATE TABLE Subject_2020201040 (SubjectID int NOT NULL, PK_ID_Student int NOT NULL, ...)	0 row(s) affected	0.011 sec

10. insertion into table

The screenshot shows the SQL Studio interface with a query window containing the following SQL code:

```
INSERT INTO Students_2020201040 (ID_Student, No/Number, FirstName, LastName, Age)
VALUES (1234, 2020201040, 'Sal Arash', 'Nayir', 20);
```

The Action Output pane at the bottom shows the execution results of the queries:

Time	Action	Response	Duration / Fetch Time
22:17:14	CREATE TABLE Students_2020201040 (ID_Student int NOT NULL, No/Number int NOT NULL, ...)	0 row(s) affected	0.0076 sec
22:17:16	ALTER TABLE Students_2020201040 DROP CO...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.001 sec
22:17:16	ALTER TABLE Students_2020201040 ADD COL...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
22:17:41	ALTER TABLE Students_2020201040 MODIFY...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.013 sec
22:18:16	CREATE TABLE Subject_2020201040 (Subj...	0 row(s) affected	0.018 sec
22:18:56	SELECT * FROM Students_2020201040 LIMIT...	0 row(s) returned	0.0064 sec / 0.000...
22:20:10	SELECT * FROM Students_2020201040 WHERE...	0 row(s) returned	0.00089 sec / 0.000...
22:20:22	SELECT * FROM Students_2020201040 WHERE...	0 row(s) returned	0.0018 sec / 0.000...
22:20:40	SELECT * FROM Students_2020201040 ORDER...	0 row(s) returned	0.0010 sec / 0.000...
22:20:61	SELECT * FROM Students_2020201040 ORDER...	0 row(s) returned	0.00046 sec / 0.000...
22:21:14	INSERT INTO Students_2020201040 (ID_Student, ...)	1 row(s) affected	0.0023 sec
22:21:21	SELECT * FROM Students_2020201040 WHERE...	1 row(s) returned	0.00046 sec / 0.000...

11 Selection of data from table

The screenshot shows a SQL query in a text editor with the following steps:

- 76. VALUES (1245, 20162153, 'Sai Anirudh', 'Karre', 35);
- 77. -- Selection of data from table
- 79. SELECT * FROM Students_2026201646;
- 80.
- 81. -- Filtering the single value data
- 82. SELECT * FROM Students_2026201646 WHERE RollNumber = 20162153;
- 83.
- 84. -- Filtering multi-value data

The query is executed, and the results are displayed in a table with columns: ID_Student, RollNumber, FirstName, LastName, Age, City. The results show 10 rows of student data.

Below the results, the 'Action Output' tab shows the execution plan and response for each step of the query.

12 Filtering the single value data

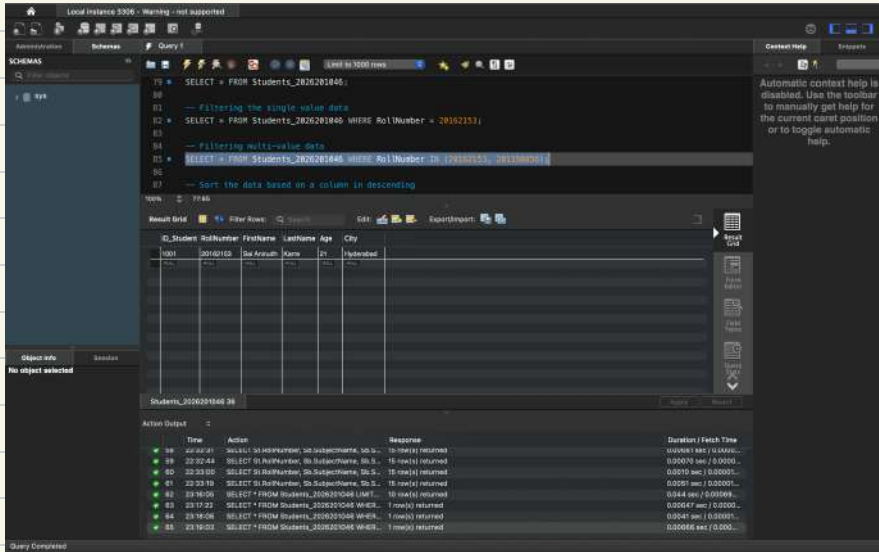
The screenshot shows a SQL query in a text editor with the following steps:

- 77. -- Selection of data from table
- 79. SELECT * FROM Students_2026201646;
- 80.
- 81. -- Filtering the single value data
- 82. SELECT * FROM Students_2026201646 WHERE RollNumber = 20162153;
- 83.
- 84. -- Filtering multi-value data
- 85. SELECT * FROM Students_2026201646 WHERE RollNumber IN (20162153, 201620864);

The query is executed, and the results are displayed in a table with columns: ID_Student, RollNumber, FirstName, LastName, Age, City. The results show 2 rows of student data.

Below the results, the 'Action Output' tab shows the execution plan and response for each step of the query.

13. Filtering multi value data



[illegible]

The screenshot displays the DBeaver SQL editor interface. The top toolbar includes icons for file operations, database connections, and query execution. The main editor area contains a SQL query with comments explaining the steps: selecting data based on a column, sorting by RollNumber DESC, sorting by RollNumber ASC, and performing a search using a filter on a table with multiple conditions using AND.

Below the editor, the 'Result Grid' shows a table with columns: ID_Student, RollNumber, FirstName, LastName, Age, and City. The table contains 10 rows of student data.

At the bottom, the 'Action Output' tab shows a log of SQL statements and their execution status, including 'row(s) returned' and 'row(s) affected'.

On the right side, a 'Context Help' panel is visible, providing information about the 'Automatic context help' feature, which can be disabled or manually triggered.

16. Pattern search using filter on a table

The screenshot shows a SQL query in a text editor with the following SQL code:

```

-- Pattern search using filter on a table with multiple condition using AND
SELECT *
FROM Students_2020201046
WHERE FirstName = 'Sal' AND RollNumber LIKE '201046';
-- Pattern search using filter on a table with multiple condition using OR

```

The query is executed, and the results are displayed in a table with columns: ID_Student, RollNumber, FirstName, LastName, Age, City. The results are empty.

The bottom panel shows the 'Action Output' table with columns: Time, Action, Response, Duration / Fetch Time. The output shows the execution of the query with the following details:

Time	Action	Response	Duration / Fetch Time
22:19:56	SELECT * FROM Students_2020201046 LIMIT...	0 rows(s) returned	0.0004 sec / 0.0004...
22:20:02	SELECT * FROM Students_2020201046 WHERE...	0 rows(s) returned	0.0004 sec / 0.0004...
22:20:07	SELECT * FROM Students_2020201046 WHERE...	0 rows(s) returned	0.0019 sec / 0.0000...
22:20:40	SELECT * FROM Students_2020201046 ORDER...	0 rows(s) returned	0.0010 sec / 0.0000...
22:21:01	INSERT INTO Students_2020201046 GO_Student...	1 row(s) affected	0.0004 sec / 0.000...
22:21:21	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0004 sec / 0.000...
22:22:30	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0014 sec / 0.0001...
22:22:30	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0018 sec / 0.0001...
22:23:06	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0007 sec / 0.000...
22:23:17	SELECT * FROM Students_2020201046 ORDER...	1 row(s) returned	0.0006 sec / 0.000...
22:23:30	SELECT * FROM Students_2020201046 ORDER...	1 row(s) returned	0.0008 sec / 0.000...
22:23:44	SELECT * FROM Students_2020201046 WHERE...	0 rows(s) returned	0.0018 sec / 0.0000...

17. Pattern search using filter with multiple condition using OR.

The screenshot shows a SQL query in a text editor with the following SQL code:

```

WHERE FirstName = 'Sal' AND RollNumber LIKE '201046';
-- Pattern search using filter on a table with multiple condition using OR
SELECT *
FROM Students_2020201046
WHERE FirstName = 'Sal' OR Age = 31;

```

The query is executed, and the results are displayed in a table with columns: ID_Student, RollNumber, FirstName, LastName, Age, City. The results are empty.

The bottom panel shows the 'Action Output' table with columns: Time, Action, Response, Duration / Fetch Time. The output shows the execution of the query with the following details:

Time	Action	Response	Duration / Fetch Time
22:20:19	SELECT * FROM Students_2020201046 WHERE...	0 row(s) returned	0.0006 sec / 0.000...
22:20:27	SELECT * FROM Students_2020201046 WHERE...	0 row(s) returned	0.0019 sec / 0.0000...
22:20:40	SELECT * FROM Students_2020201046 ORDER...	0 rows(s) returned	0.0012 sec / 0.0000...
22:21:01	SELECT * FROM Students_2020201046 ORDER...	0 rows(s) returned	0.0004 sec / 0.000...
22:21:14	INSERT INTO Students_2020201046 GO_Student...	1 row(s) affected	0.0004 sec / 0.000...
22:21:31	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0004 sec / 0.000...
22:22:30	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0014 sec / 0.0001...
22:22:30	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0018 sec / 0.0001...
22:23:17	SELECT * FROM Students_2020201046 ORDER...	1 row(s) returned	0.0006 sec / 0.000...
22:23:30	SELECT * FROM Students_2020201046 ORDER...	1 row(s) returned	0.0008 sec / 0.000...
22:23:44	SELECT * FROM Students_2020201046 WHERE...	0 rows(s) returned	0.0018 sec / 0.0000...
22:24:00	SELECT * FROM Students_2020201046 WHERE...	0 rows(s) returned	0.0018 sec / 0.0000...

18. Filtering using NOT

(I)

The screenshot shows a SQL query in the 'Query 1' editor. The query is as follows:

```
-- Filter using NOT ~ different variants and wildcard
103 SELECT * FROM Students_2026201046 WHERE LastName = 'Anirudh';
104 SELECT * FROM Students_2026201046 WHERE FirstName NOT LIKE 'Sai';
105 SELECT * FROM Students_2026201046 WHERE RollNumber NOT BETWEEN 20162151 AND 20162169;
106 SELECT * FROM Students_2026201046 WHERE RollNumber NOT IN (20162151, 20162169);
107 SELECT * FROM Students_2026201046 WHERE NOT Age > 17;
108
-- Updating data using a Filter
109
110
111 UPDATE Students_2026201046
```

The 'Result Grid' displays the following data:

ID_Student	RollNumber	FirstName	LastName	Age	City
1001	20162153	Ravi	Kumar	20	Hyderabad
1002	20162154	Ravi	Kumar	22	Chennai
1003	20162155	Ravi	Kumar	20	Delhi
1004	20162156	Anjali	Ravi	23	Hyderabad
1005	20162157	Ravi	Kumar	22	Hyderabad
1006	20162158	Ravi	Kumar	19	Chennai
1007	20162159	Ravi	Kumar	22	Mumbai
1008	20162160	Ravi	Kumar	20	Hyderabad
1009	20162161	Vikram	Singh	24	Delhi
1010	20162162	Ravi	Kumar	20	Chennai

The 'Action Output' table shows the execution of the query:

Time	Action	Response	Duration / Fetch Time
66	23:19:42	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
67	23:19:43	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
68	23:19:44	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
69	23:19:45	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
70	23:19:46	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
71	23:19:47	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
72	23:19:48	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec

(II)

The screenshot shows a SQL query in the 'Query 1' editor. The query is as follows:

```
-- Filter using NOT ~ different variants and wildcard
103 SELECT * FROM Students_2026201046 WHERE LastName = 'Anirudh';
104 SELECT * FROM Students_2026201046 WHERE FirstName NOT LIKE 'Sai';
105 SELECT * FROM Students_2026201046 WHERE RollNumber NOT BETWEEN 20162151 AND 20162169;
106 SELECT * FROM Students_2026201046 WHERE RollNumber NOT IN (20162151, 20162169);
107 SELECT * FROM Students_2026201046 WHERE NOT Age > 17;
108
-- Updating data using a Filter
109
110
111 UPDATE Students_2026201046
```

The 'Result Grid' displays the following data:

ID_Student	RollNumber	FirstName	LastName	Age	City
1005	20162154	Ravi	Kumar	20	Chennai
1006	20162155	Ravi	Kumar	22	Delhi
1007	20162156	Anjali	Ravi	23	Hyderabad
1008	20162157	Ravi	Kumar	22	Hyderabad
1009	20162158	Ravi	Kumar	19	Chennai
1010	20162159	Ravi	Kumar	22	Mumbai
1011	20162160	Ravi	Kumar	20	Hyderabad
1012	20162161	Vikram	Singh	24	Delhi
1013	20162162	Ravi	Kumar	20	Chennai

The 'Action Output' table shows the execution of the query:

Time	Action	Response	Duration / Fetch Time
66	23:19:42	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
67	23:19:43	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
68	23:19:44	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
69	23:19:45	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
70	23:19:46	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
71	23:19:47	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec
72	23:19:48	SELECT * FROM Students_2026201046 WHERE...	0.00046 sec / 0.00046 sec

Local Instance 3306 - Warning - not supported

Schemas

Query 1

Limit to 1000 rows

103 -- Filter using NOT - different variants and wildcard
104 * SELECT * FROM Students_2026201046 WHERE NOT LastName = 'Amirrah';
105 * SELECT * FROM Students_2026201046 WHERE FirstName NOT LIKE 'Saiv';
106 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT BETWEEN 20162153 AND 20162169;
107 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT IN (20162153, 20162169);
108 * SELECT * FROM Students_2026201046 WHERE NOT Age > 17;
109

Result Grid

0 rows returned

Students_2026201046 10

Object Info

No object selected

Action Output

Time	Action	Response	Duration / Fetch Time
22 22:14	SELECT * FROM Students_2026201046 ORDER...	0 row(s) returned	0.00445 sec / 0.000...
22 22:14	UPDATE INTO Students_2016201046 IS Stud...	1 row(s) affected	0.0033 sec
22 22:14	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00445 sec / 0.000...
22 22:35	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.004 sec / 0.00001...
22 22:50	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.0016 sec / 0.0000...
22 22:59	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00087 sec / 0.000...
22 22:57	SELECT * FROM Students_2026201046 ORDER...	1 row(s) returned	0.00565 sec / 0.000...
22 22:59	SELECT * FROM Students_2026201046 ORDER...	1 row(s) returned	0.00565 sec / 0.000...
22 22:54	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0019 sec / 0.0000...
22 22:40	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00080 sec / 0.000...
22 22:44	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00091 sec / 0.0000...
22 22:40	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0017 sec / 0.0000...
22 22:43	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00087 sec / 0.000...

Query Completed

Local Instance 3306 - Warning - not supported

Schemas

Query 1

Limit to 1000 rows

103 -- Filter using NOT - different variants and wildcard
104 * SELECT * FROM Students_2026201046 WHERE NOT LastName = 'Amirrah';
105 * SELECT * FROM Students_2026201046 WHERE FirstName NOT LIKE 'Saiv';
106 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT BETWEEN 20162153 AND 20162169;
107 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT IN (20162153, 20162169);
108 * SELECT * FROM Students_2026201046 WHERE NOT Age > 17;
109
110 -- Updating data using a Filter
111 * UPDATE Students_2026201046
112
113

Result Grid

0 rows returned

Students_2026201046 48

Object Info

No object selected

Action Output

Time	Action	Response	Duration / Fetch Time
22 22:22	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00445 sec / 0.0000...
22 22:23	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00086 sec / 0.000...
22 22:44	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00089 sec / 0.000...
22 22:40	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00072 sec / 0.000...
22 22:41	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00048 sec / 0.000...
22 22:59	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0025 sec / 0.0000...
22 22:57	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00015 sec / 0.000...
22 22:59	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00075 sec / 0.000...

Query Completed

Local Instance 5306 - Warning - not supported

Administration Schemas Query 1

SCHEMAS

Filter using NOT - Different variants and wildcard

```

103 -- Filter using NOT - Different variants and wildcard
104 * SELECT * FROM Students_2026201046 WHERE NOT LastName = 'Anirudh';
105 * SELECT * FROM Students_2026201046 WHERE FirstName NOT LIKE 'Sai';
106 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT BETWEEN 20162153 AND 20162160;
107 * SELECT * FROM Students_2026201046 WHERE RollNumber NOT IN (20162153, 20162160);
108 * SELECT * FROM Students_2026201046 WHERE NOT Age < 19;
109
100% 0 0:00:00

```

Result Grid

ID_Student	RollNumber	FirstName	LastName	Age	City
Students_2026201046 18					

Object Info

No object selected

Session Action Output

Time	Action	Response	Duration / Fetch Time
31 22:21:21	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00045 sec / 0.000...
32 22:22:36	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.0014 sec / 0.0001...
33 22:22:40	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.0016 sec / 0.0001...
34 22:23:08	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00087 sec / 0.000...
35 22:23:17	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00086 sec / 0.000...
36 22:23:30	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00089 sec / 0.000...
37 22:23:44	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0019 sec / 0.0003...
38 22:24:00	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00082 sec / 0.000...
39 22:24:16	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00081 sec / 0.000...
40 22:24:30	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0017 sec / 0.0000...
41 22:24:49	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00087 sec / 0.000...
42 22:25:05	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00086 sec / 0.000...
43 22:25:16	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00078 sec / 0.000...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

19. Update data using condition

Local Instance 5306 - Warning - not supported

Administration Schemas Query 1

SCHEMAS

Filter using NOT - Different variants and wildcard

```

110 -- Updating data using a Filter
111 * UPDATE Students_2026201046
112 SET Age = 23, City = 'Hyderabad'
113 WHERE RollNumber < 20162153;
114
115 -- Deletion of data for specific filter
116 * DELETE FROM Students_2026201046 WHERE RollNumber < 20162153;
117
118 -- Aggregate Functions MIN(), MAX(), COUNT(), SUM(), AVG()
119 -- (Original queries used a table called 'Subjects' that was never created -
120 -- standardized to Subject_2026201046)
121 * SELECT MIN(Marks) FROM Subject_2026201046 WHERE RollNumber < 20162153;
122 * SELECT MAX(Marks) FROM Subject_2026201046 WHERE RollNumber < 20162153;
123 * SELECT SUM(Marks) FROM Subject_2026201046 WHERE RollNumber < 20162153 AND Semester = 2;
124 * SELECT AVG(Marks) FROM Subject_2026201046 WHERE RollNumber < 20162153 AND Semester IN (1, 2, 3);
125 * SELECT COUNT(RollNumber) FROM Subject_2026201046
126 WHERE Marks < 80 AND SubjectName = 'ISS' AND Semester = 2;
127
128 -- Group By & Having to be used with Aggregate Function for summary
129 * SELECT SubjectName, COUNT(RollNumber) AS StudentCount
130 FROM
131
100% 0 0:00:00

```

Result Grid

ID_Student	RollNumber	FirstName	LastName	Age	City
Students_2026201046 18					

Object Info

No object selected

Session Action Output

Time	Action	Response	Duration / Fetch Time
31 22:22:36	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.0014 sec / 0.0001...
32 22:22:40	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.0016 sec / 0.0001...
33 22:23:08	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00087 sec / 0.000...
34 22:23:17	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00086 sec / 0.000...
35 22:23:30	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00089 sec / 0.000...
36 22:23:44	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0019 sec / 0.0003...
37 22:24:00	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00082 sec / 0.000...
38 22:24:16	SELECT * FROM Students_2026201046 WHERE...	1 row(s) returned	0.00081 sec / 0.000...
39 22:24:30	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.0017 sec / 0.0000...
40 22:24:49	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00087 sec / 0.000...
41 22:25:05	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00086 sec / 0.000...
42 22:25:16	SELECT * FROM Students_2026201046 WHERE...	0 row(s) returned	0.00078 sec / 0.000...
43 22:25:30	UPDATE Students_2026201046 SET Age = 23;	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.018 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

20. Deleting Record using specific condition

Local Instance 3300 - Warning - not supported

Administration Schemas Query 1

119 -- Updating data using a Filter
 111 * UPDATE Students_2020201046
 112 SET Age = 23, City = 'Hyderabad'
 113 WHERE RollNumber = 20102153;
 114
 115 -- Deletion of data for specific filter
 116 * DELETE FROM Students_2020201046 WHERE RollNumber = 20102153;
 117
 118 -- Aggregate functions ROW(), MAX(), COUNT(), SUM(), AVG()
 119 -- (original queries used a table called "Subjects" that was never created -
 120 -- standardized to Subject_2020201046)
 121 * SELECT MIN(Marks) FROM Subject_2020201046 WHERE RollNumber = 20102153;
 122 * SELECT MAX(Marks) FROM Subject_2020201046 WHERE RollNumber = 20102153;
 123 * SELECT SUM(Marks) FROM Subject_2020201046 WHERE RollNumber = 20102153 AND Semester = 2;
 124 * SELECT AVG(Marks) FROM Subject_2020201046 WHERE RollNumber = 20102153 AND Semester IN (1, 2, 3);
 125 * SELECT COUNT(RollNumber) FROM Subject_2020201046
 126 WHERE Marks > 80 AND SubjectName = 'ISS' AND Semester = 2;
 127
 128 -- Group by & Having to be used with Aggregate Function for summary
 129 * SELECT SubjectName, COUNT(RollNumber) AS StudentCount
 130 FROM Subject_2020201046
 131 GROUP BY SubjectName
 132 HAVING StudentCount > 10

Object Info: No object selected

Action Output: 0

Time	Action	Response	Duration / Fetch Time
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.0015 sec / 0.0001...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00087 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00068 sec / 0.000...
22:23:50	UPDATE Students_2020201046 SET Age = 23...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
22:23:50	DELETE FROM Students_2020201046 WHERE...	1 row(s) affected	0.0015 sec

Query Completed

21. Aggregate Functions

(I) MIN

Local Instance 3300 - Warning - not supported

Administration Schemas Query 1

142 (14, 1801, 'Networks', 20102153, 89, 21)
 143 (15, 1802, 'ISS', 20102154, 98, 21)
 144
 145 -- Aggregate functions ROW(), MAX(), COUNT(), SUM(), AVG()
 146 -- (original queries used a table called "Subjects" that was never created -
 147 -- standardized to Subject_2020201046)
 148 * SELECT MIN(Marks) FROM Subject_2020201046 WHERE RollNumber = 20102153;
 149
 150 -- Group by & Having to be used with Aggregate Function for summary
 151 * SELECT SubjectName, COUNT(RollNumber) AS StudentCount
 152 FROM Subject_2020201046
 153 GROUP BY SubjectName
 154 HAVING StudentCount > 10

Object Info: No object selected

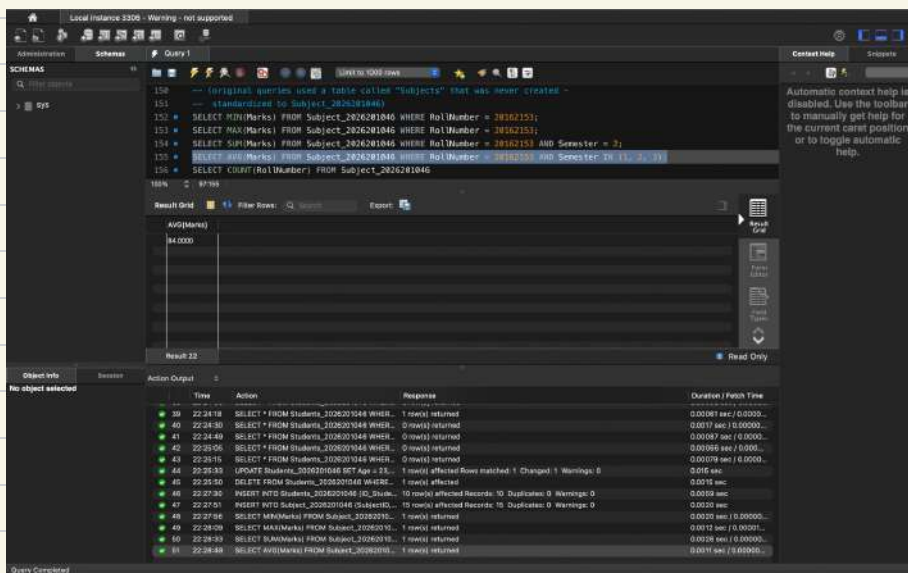
Action Output: 0

Time	Action	Response	Duration / Fetch Time
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	SELECT * FROM Students_2020201046 WHERE...	1 row(s) returned	0.00088 sec / 0.000...
22:23:50	UPDATE Students_2020201046 SET Age = 23...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
22:23:50	DELETE FROM Students_2020201046 WHERE...	1 row(s) affected	0.0015 sec
22:23:50	INSERT INTO Students_2020201046 (SubjectID...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.0069 sec
22:23:50	SELECT MIN(Marks) FROM Subject_2020201046...	1 row(s) returned	0.0015 sec

Query Completed

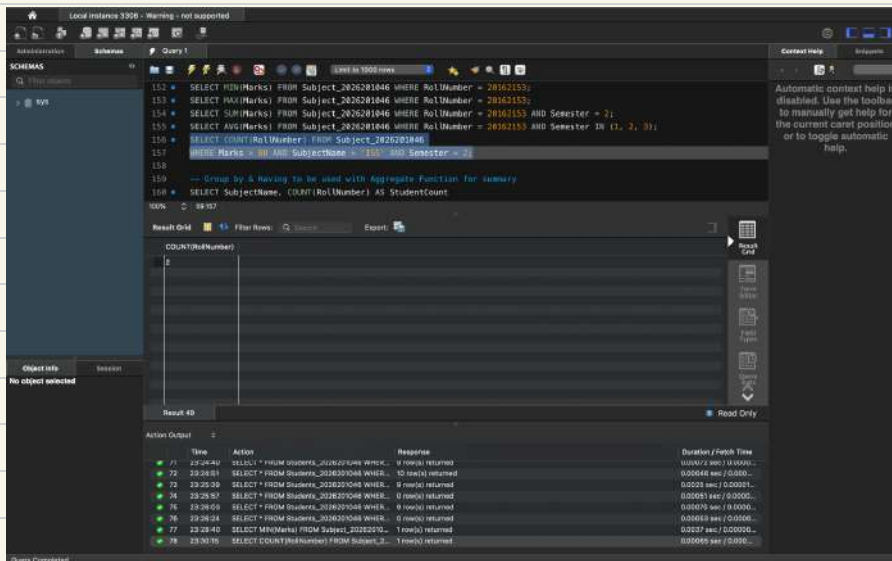
(IV)

AVG



(V)

COUNT



22 Group by & Having used with aggregate functions

The screenshot shows the SQL Server Enterprise Manager interface. The query editor contains the following SQL code:

```
-- Group By & Having to be used with Aggregate Function for a summary
SELECT SubjectName, COUNT(RollNumber) AS StudentCount
FROM Subject_2026201046
GROUP BY SubjectName
HAVING COUNT(RollNumber) > 5
```

The Results pane shows a table with 25 rows and 2 columns: SubjectName and StudentCount. The table is currently empty.

The Action Output pane shows the execution plan for the query. The query was completed successfully, returning 25 rows.

23. Column Alias

The screenshot shows the SQL Server Enterprise Manager interface. The query editor contains the following SQL code:

```
SELECT COUNT(RollNumber) > 5
-- Column Alias
SELECT RollNumber AS ID FROM Students_2026201046
-- ORDER - inner
SELECT S1.RollNumber, S1.SubjectName, S1.Semester
```

The Results pane shows a table with 25 rows and 1 column: ID. The table is currently empty.

The Action Output pane shows the execution plan for the query. The query was completed successfully, returning 25 rows.

25. Left Join of Students & Subjects Table

Local Instance XDBE - Warning: not supported

Administration | Schema | Query | Contact Help | Export

SCHEMAS | sys

Query Editor

```
169 SELECT St.RollNumber, Sb.SubjectName, Sb.Semester
170 FROM Students_2026201046 AS St
171 INNER JOIN Subject_2026201046 AS Sb ON St.St.RollNumber = Sb.RollNumber;
172
173 -- 30/04 - left
174 SELECT St.RollNumber, Sb.SubjectName, Sb.Semester
175 FROM Students_2026201046 AS St
176 LEFT JOIN Subject_2026201046 AS Sb ON St.St.RollNumber = Sb.RollNumber;
177
```

Result Grid | Filter Rows | Export

RollNumber	SubjectName	Semester
20480163	Networks	2
20480163	OS	1
20480163	DBMS	1
20480164	ISS	2
20480164	OS	1
20480164	DBMS	1
20480165	ISS	2
20480165	DBMS	2
20480166	Networks	2
20480166	DBMS	1
20480166	OS	3
20480166	ISS	3
20480169	Networks	2
20480169	DBMS	1
20480169	OS	1

Object Info | Details

No object selected

Action Output

Time	Action	Response	Duration / Elapsed Time
22:28:53	SELECT * FROM STUDENTS WHERE SUBJECT_2026201046...	1 row(s) returned	0.0000 sec / 0.0000 sec
22:28:54	SELECT AVG(Networks) FROM Student_2026201046...	1 row(s) returned	0.0011 sec / 0.0000 sec
22:28:55	SELECT COUNT(RollNumber) FROM Student_2...	1 row(s) returned	0.0033 sec / 0.0000 sec
22:30:16	SELECT COUNT(RollNumber) FROM Subject_2...	1 row(s) returned	0.0000 sec / 0.0000 sec
22:30:23	SELECT SubjectName, COUNT(RollNumber) AS...	0 row(s) returned	0.0056 sec / 0.0000 sec
22:30:42	SELECT RollNumber AS St FROM Students_20...	10 row(s) returned	0.0007 sec / 0.0000 sec
22:31:10	SELECT St.RollNumber, Sb.SubjectName, Sb...	16 row(s) returned	0.0016 sec / 0.0000 sec
22:31:43	SELECT St.RollNumber, Sb.SubjectName, Sb...	16 row(s) returned	0.0031 sec / 0.0000 sec

Automatic contact help is disabled. Use the toolbar manually get help for the current caret position or to toggle automatic help.

Query Completed

